Power Engineering and Engineering Thermophysics
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# Combined Effects of Hybrid Nanofluids, Metal Foam, and Thermal Radiation on Phase Change Material Solidification in Cold Thermal Energy Storage

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**Received:** 06-10-2026 **Revised:** 07-17-2026 **Accepted:** 08-02-2026

**Citation:** J. N. Abu-Hamdeh, "Combined effects of hybrid nanofluids, metal foam, and thermal radiation on phase change material solidification in cold thermal energy storage," *Power Eng. Eng. Thermophys.*, vol. 5, no. 3, pp. 223–233, 2026. https://doi.org/10.56578/peet050304.



**Abstract:** The performance of cold thermal energy storage systems is often limited by the low thermal conductivity of phase change materials (PCMs), which delays solidification and reduces charging efficiency. In the present study, the synergistic effects of hybrid nanofluids, porous metal foam, and thermal radiation on the solidification behavior of a PCM-based cold thermal energy storage unit incorporating elliptical and triangular cooling boundaries were numerically investigated. A transient numerical model was developed using the Galerkin finite element method and was coupled with an implicit time-integration scheme and adaptive mesh refinement to accurately resolve temperature evolution and the moving solid–liquid interface during the freezing process. The numerical framework was validated against benchmark results available in the literature, and excellent agreement was achieved. It was found that the addition of hybrid nanoparticles reduced the total solidification time by approximately 5.19%. When thermal radiation was incorporated, the freezing duration was further shortened by nearly 34.35%. The most pronounced enhancement was obtained through the incorporation of porous metal foam, for which the solidification time was reduced by approximately 75.25% as a result of the substantial augmentation of conductive heat transfer pathways within the PCM. Under the combined application of all enhancement mechanisms, the total freezing time was reduced by up to 84.59% relative to the baseline configuration. These findings demonstrate that the integration of porous structures, radiative cooling, and hybrid nanofluids represents an effective strategy for overcoming the thermal limitations of conventional PCM-based cold thermal energy storage systems and provides valuable design guidance for the development of high-efficiency thermal energy storage technologies in industrial cooling and sustainable energy systems.

**Keywords:** Cold thermal energy storage; Phase change material; Hybrid nanofluid; Metal foam; Solidification enhancement; Thermal radiation; Galerkin finite element method

## 1 Introduction

The freezing process within a porous container, especially when hybrid nanoparticles are incorporated, plays a vital role in improving the thermal management and efficiency of phase change materials (PCMs). The porous container's structure increases the surface area, facilitating more effective heat transfer between the solidifying PCM and its surrounding environment [1–3]. The inclusion of hybrid nanoparticles in the PCM enhances the material's heat conduction properties, resulting in faster solidification rates. These nanoparticles, often made from materials like metal oxides or carbon-based compounds, significantly improve the overall conductivity, allowing more efficient heat transfer across the entire system [4–6]. During solidification, this leads to a more even temperature distribution within the PCM, minimizing the development of thermal gradients that could hinder the solidification process. The amalgamation of the porous structure and hybrid nanoparticles speeds up the phase change, allowing for faster and more controlled solidification. This is especially advantageous in applications where fast cooling or freezing is crucial, such as thermal energy storage, refrigeration, and waste heat recovery systems. By enhancing the solidification process, the overall performance can be greatly improved, offering more reliable and cost-effective energy management solutions [7–9]. Al-Hinti et al. [10] employed PCMs in a solar water heater system and developed an experimental setup. Their findings demonstrated that the system was able to provide hot water at 55 °C during the daytime.

Al-Aasam et al. [11] investigated a nanofluid-based solar panel integrated with a paraffin container. Their findings revealed that the PCM-photovoltaic/thermal unit with a twisted tube and 0.6% nanofluid achieved the greatest performance (94.31%). Ibrahim et al. [12] enhanced energy storage productivity in Iraq by incorporating hybrid titanium dioxide-magnesium oxide ($TiO_2$-MgO) nanoparticles. This combination resulted in a maximum thermal conductivity increase of 24.92%, demonstrating its potential for improving the efficiency of solar energy applications. Shaikh and Lafdi [13] introduced a new festoon-type storage configuration for multiple PCM placements. Their study demonstrated a 30% improvement in the charging rate of the festoon configuration compared to the single-stage festoon design. Al-Waeli et al. [14] derived a new model for predicting the behavior of solar panels in existence of nano-enhanced PCMs. The model was able to accurately predict the temperature distribution, with the thermal efficiency reaching 72%. Mirparizi [15] developed an innovative cold thermal energy storage system incorporating porous metal foam to improve the solidification process. Their numerical investigation demonstrated that the presence of porous media significantly enhanced heat transfer, resulting in a considerable reduction in freezing time due to the increased effective thermal conductivity of the storage medium. Rothan [16] carried out a numerical study to evaluate the effect of nanoparticle incorporation on the freezing behavior of PCMs. The research focused on the influence of nanoparticle concentration on solidification performance and showed that increasing the nanoparticle volume fraction accelerated the freezing process by enhancing the thermal conductivity of the PCM. Alturaihi and Abd Ali [17] numerically examined the application of PCMs for the thermal regulation of photovoltaic panels. Their results revealed that integrating a PCM layer beneath the solar panel effectively reduced the module operating temperature, thereby improving the electrical conversion efficiency and enhancing the overall performance of the photovoltaic system. Almohsen [18] proposed an advanced thermal energy storage configuration incorporating nanoparticles to improve the solidification characteristics of PCMs. Employing the Galerkin finite element method, the study investigated the freezing behavior under various design conditions and quantified the corresponding solidification times, demonstrating the effectiveness of nanoparticle-enhanced thermal energy storage systems.

In recent years, there has been growing interest in developing more efficient cold energy storage systems to address the increasing global demand for sustainable energy solutions. In this context, the integration of innovative materials and advanced geometries has proven to significantly enhance the performance of such systems. The current study introduces a novel methodology by employing a container design with triangular and elliptical cold surfaces, which provides a unique geometry for enhancing heat transfer and freezing rates. Unlike traditional methods that use water as a PCM, this work utilizes a hybrid nanomaterial to optimize the thermal conductivity of the system. The incorporation of foam further improves the freezing rate by enhancing conduction. The innovation of the current study lies in the combination of these three techniques—hybrid nanomaterials, porous foam, and the advanced container design—with a carefully derived mathematical model to capture their interactions. By neglecting the effect of gravity and using a homogeneous nanofluid mixture, the model simplifies the system while retaining its essential physical characteristics. The equations governing the system are solved using the Galerkin method, a widely recognized technique for handling complex geometries, making this method both efficient and effective. Compared to previous studies, which often focus on individual techniques or simpler geometries, this research represents a significant advancement by integrating multiple methods to enhance cold energy storage units. The findings highlight the substantial impact of these combined techniques on reducing freezing times, demonstrating that this approach could be highly beneficial for future cold storage systems. This work fills an important gap in the literature by offering a comprehensive analysis of the interaction between these factors and providing a new direction for research into more efficient and sustainable cold energy storage solutions.

## 2 Development of a Numerical Approach for Modeling the Cold Storage Process

This study employed a novel container design with two triangular and elliptical cold surfaces for cold energy storage (Figure 1). The tank was filled with a hybrid nanomaterial instead of conventional water, and porous foam was incorporated to enhance the freezing rate. To derive the governing equations, the impact of gravity force was neglected, and a homogeneous mixture model was used for the nanofluid. When calculating the properties of the porous media, the impact of foam porosity was considered. To visualize the effects of the three techniques (nanofluid, radiation, and porous foam), contour plots of scalar quantities were generated for various scenarios. Additionally, the solid front progression, average scalar values, and total completion time were compared across different cases to highlight the improvements achieved by each technique. For the solidification process modeling, two key scalars must be derived at each node, as outlined in the governing equations. These scalars provide the essential data for determining the state of the system as it transitions from liquid to solid [19].

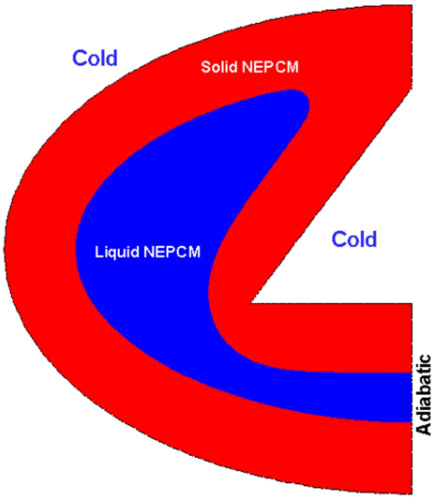


**Figure 1.** A container with elliptic and triangular cold surfaces for cold energy storage
Note: NEPCM = nano-enhanced phase change material.

To describe the transient heat transfer during the solidification process, the governing energy equation, including the effects of heat conduction, latent heat, and thermal radiation, is expressed as follows:

$$
\begin{gathered}
(\rho C_{\mathrm{p}})_{\mathrm{eff}} \frac{\mathrm{d}T}{\mathrm{d}t} = k_{\mathrm{eff}} \left( \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial x^2} \right) + (L\rho)_{\mathrm{NEPCM}} \frac{\partial S}{\partial t} + \frac{\partial q_{\mathrm{r},y}}{\partial y} + \frac{\partial q_{\mathrm{r},x}}{\partial x}, \\
q_{\mathrm{r},y} = -\frac{4\sigma}{3\beta_{\mathrm{R}}} \frac{\partial T^4}{\partial y}, \quad q_{\mathrm{r},x} = -\frac{4\sigma}{3\beta_{\mathrm{R}}} \frac{\partial T^4}{\partial x}, \quad T^4 \cong 4T_0^3 T - 3T_0^4, \quad Rd = \frac{16\sigma T_0^3}{\beta_{\mathrm{R}} k_{\mathrm{f}}}
\end{gathered} \tag{1}
$$

where, $T$ is the temperature, $t$ is the time, $x$ and $y$ are the spatial coordinates, $(\rho C_{\mathrm{p}})_{\mathrm{eff}}$ is the effective volumetric heat capacity, $k_{\mathrm{eff}}$ is the effective thermal conductivity, $L$ is the latent heat, $S$ is the solid fraction, and $q_{r,x}$ and $q_{r,y}$ denote the radiative heat flux components in the $x$- and $y$-directions, respectively. $Rd$ represents the radiation parameter.

The local solid fraction is determined as a function of temperature using the following piecewise relation:

$$
\begin{cases}
T > (T_{\mathrm{m}} + T_0) \Rightarrow S = 0, \\
(-T_0 + T_{\mathrm{m}}) < T < (T_0 + T_{\mathrm{m}}) \Rightarrow S = (-T + 0.5T_0 + T_{\mathrm{m}})/T_0, \\
T < (T_{\mathrm{m}} - T_0) \Rightarrow S = 1
\end{cases} \tag{2}
$$

where, $S = 0$ represents the liquid phase and $S = 1$ represents the fully solid phase, while $T_{\mathrm{m}}$ denotes the phase-change temperature.

The effective thermophysical properties of the porous medium are calculated by accounting for the contributions of the NEPCM and the solid foam matrix as follows:

$$
\begin{aligned}
(\rho C_{\mathrm{p}})_{\mathrm{eff}} &= \gamma(\rho C_{\mathrm{p}})_{\mathrm{NEPCM}} + (1-\gamma)(\rho C_{\mathrm{p}})_{\mathrm{GI}} \\
k_{\mathrm{eff}} &= \gamma k_{\mathrm{NEPCM}} + (1-\gamma) k_{\mathrm{GI}}
\end{aligned} \tag{3}
$$

where, $\gamma$ denotes the porosity of the metal foam, while the subscripts "eff" and "NEPCM" refer to the effective property and nano-enhanced phase change material, respectively.

The thermophysical properties of the water-based PCM were updated according to the volume fractions of the additives to accurately represent the behavior of the hybrid nanofluid during freezing [15]. Accordingly, the density of the nano-enhanced phase change material was evaluated based on the volume fractions of the base PCM and the two nanoparticle components as follows:

$$
\begin{aligned}
\rho_{\mathrm{NEPCM}} &= (1-\phi_{\mathrm{hnf}})\rho_{\mathrm{PCM}} + \phi_{\mathrm{Al_2O_3}}\rho_{\mathrm{Al_2O_3}} + \phi_{\mathrm{CuO}}\rho_{\mathrm{CuO}}, \\
\phi_{\mathrm{hnf}} &= \phi_{\mathrm{Al_2O_3}} + \phi_{\mathrm{CuO}}
\end{aligned} \tag{4}
$$

where, $\rho_{\mathrm{NEPCM}}$ is the density of the nano-enhanced PCM, $\phi_{\mathrm{hnf}}$ is the total volume fraction of the hybrid nanoparticles, and $\phi_{\mathrm{Al_2O_3}}$ and $\phi_{\mathrm{CuO}}$ are the volume fractions of $Al_2O_3$ and CuO nanoparticles, respectively.

Similarly, the effective volumetric heat capacity of the NEPCM is determined from the contributions of the base PCM and nanoparticles as follows:

$$
(\rho C_{\mathrm{p}})_{\mathrm{NEPCM}} = (1-\phi_{\mathrm{hnf}})(\rho C_{\mathrm{p}})_{\mathrm{PCM}} + \phi_{\mathrm{Al_2O_3}}(\rho C_{\mathrm{p}})_{\mathrm{Al_2O_3}} + \phi_{\mathrm{CuO}}(\rho C_{\mathrm{p}})_{\mathrm{CuO}} \tag{5}
$$

where, $C_{\mathrm{p}}$ denotes the specific heat capacity.

The effective thermal conductivity of the NEPCM is then calculated as follows:

$$\left(\frac{k_{\mathrm{NEPCM}}}{k_{\mathrm{PCM}}}\right)=\left[\frac{\wp(1+2\phi_{\mathrm{hnf}})+2k_{\mathrm{PCM}}\phi_{\mathrm{hnf}}(1-\phi_{\mathrm{hnf}})}{\wp(1-\phi_{\mathrm{hnf}})+2k_{\mathrm{PCM}}\phi_{\mathrm{hnf}}(1-\phi_{\mathrm{hnf}})}\right]$$
$$\wp=\phi_{\mathrm{Al_2O_3}}k_{\mathrm{Al_2O_3}}+\phi_{\mathrm{CuO}}k_{\mathrm{CuO}} \tag{6}$$

where, $k_{\mathrm{NEPCM}}$ and $k_{\mathrm{PCM}}$ denote the thermal conductivities of the nano-enhanced PCM and base PCM, respectively, while $k_{\mathrm{Al_2O_3}}$ and $k_{\mathrm{CuO}}$ represent the thermal conductivities of the corresponding nanoparticles.

The grid generation was based on triangular elements, and a denser grid was applied near the solid front, with the grid density adapting as the solidification progressed. This numerical approach, initially used by Sheikholeslami [20] for energy storage systems, has proven effective for modeling complex systems such as the cold energy storage system in this study. By leveraging this established method, the present work builds on previous advancements to provide a comprehensive solution for the system's thermal dynamics.

## 3 Results and Discussion

In this study, a novel container design featuring two cold surfaces, one triangular and one elliptical, was offered for cold energy storage. The impact of foam porosity on the properties of the porous media was also considered in the calculations. The equations governing the heat transfer and phase change processes were solved using the Galerkin method. To assess the effectiveness of the three techniques—hybrid nanomaterials, porous foam, and the unique container shape—contour plots for key scalar variables were provided. These plots exemplified the progress of temperature and solid fraction over time, highlighting the impact of each technique on the freezing process. Additionally, a comprehensive comparison was made between various cases, focusing on the solid front progression, average scalar values, and the overall freezing time. The results revealed significant improvements in freezing efficiency, particularly with the integration of porous foam and the use of hybrid nanomaterials. The completion time was notably reduced, showcasing the advantages of this innovative approach in cold energy storage systems. The novelty of this study lies in the combination of several advanced techniques—such as the use of hybrid nanomaterials, the introduction of porous foam to enhance conduction, and the implementation of a novel container design with triangular and elliptical cold surfaces.

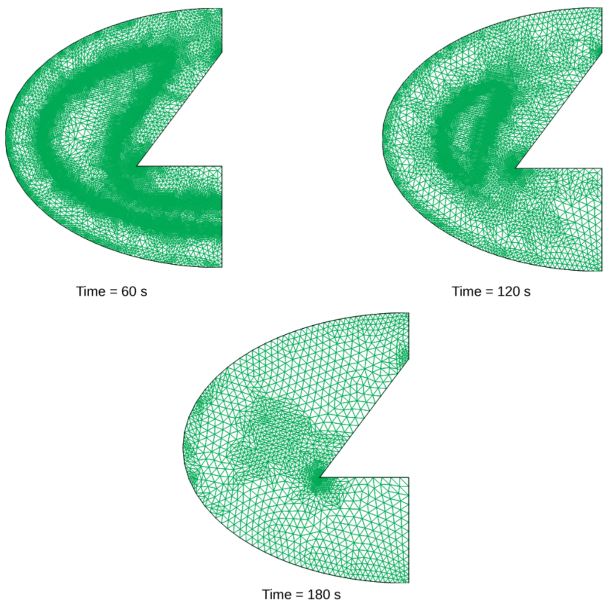


**Figure 2.** Grid generation for various time stages ($\phi_{\mathrm{hnf}}$ = 0.02, $\gamma$ = 0.95, $Rd$ = 1)

Note: $\phi_{\mathrm{hnf}}$ is the hybrid nanoparticle total volume fraction; $\gamma$ is the porosity of the porous metal foam; $Rd$ = the radiation parameter.

Figure 2 exemplifies the mesh configuration used in this study. It shows the use of various mesh shapes that adapt over time. This adaptive grid approach, which adjusts according to the position of the solid front, enhances the modeling accuracy, ensuring more precise results as the freezing process progresses. Figure 3 demonstrates the validation of the numerical code used to predict the behavior of the cold storage unit. The comparison with the study by Sheikholeslami [20] reveals that the outputs from the current code align well with prior results, indicating its accuracy. This validation confirms that the numerical model can effectively simulate the behavior of the system under investigation.

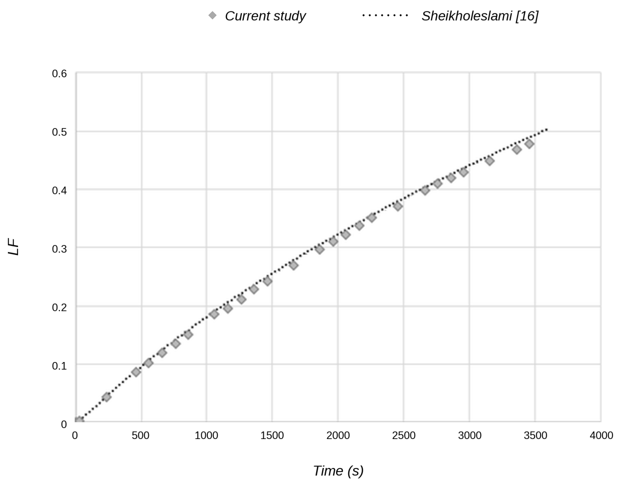


**Figure 3.** Deviation between the present results and those reported by Sheikholeslami [20]

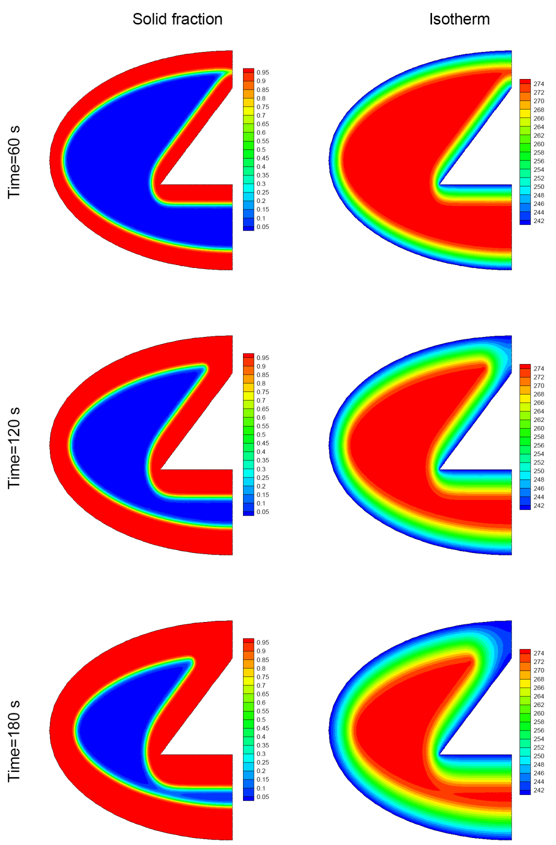


**Figure 4.** Changes in contours during freezing for the case of $\phi_{\mathrm{hnf}} = 0$, $Rd = 1$, $\gamma = 1$

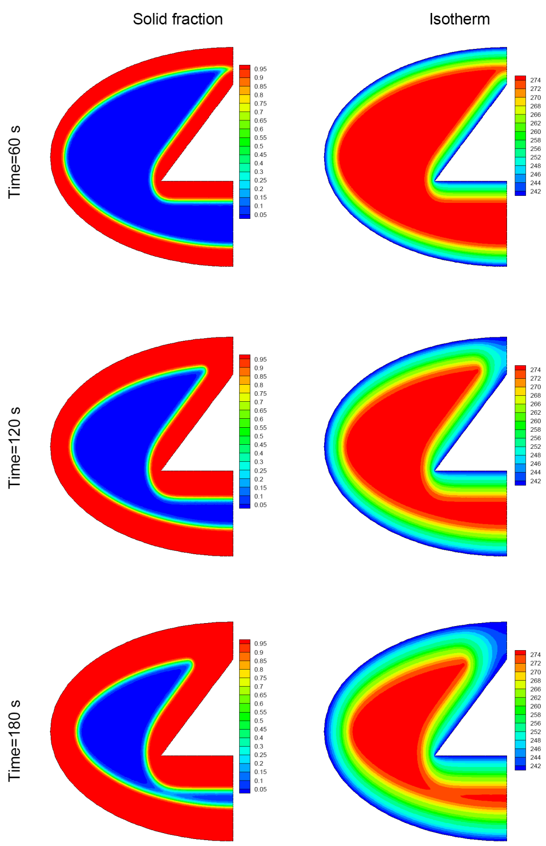


**Figure 5.** Changes in contours during freezing for the case of $\phi_{\mathrm{hnf}} = 0.02$, $Rd = 1$, $\gamma = 1$

Note: $\phi_{\mathrm{hnf}}$ is the hybrid nanoparticle total volume fraction; $\gamma$ is the porosity of the porous metal foam; $Rd$ = the radiation parameter.

The behavior of the PCM during the freezing process was analyzed across several cases, as depicted in Figure 4, Figure 5, Figure 6, Figure 7, and Figure 8. As time progresses, the solid front advances inward while the temperature of the domain steadily decreases. Notably, regions near the elliptic and triangular surfaces solidify more quickly than other areas, indicating the impact of the container's geometry on the freezing. Areas with higher temperatures remain in the liquid phase, which is further confirmed by the solid fraction contours. Among the various scenarios evaluated, the longest freezing time occurs in the absence of radiation, requiring 1136.82 seconds. In contrast, when all three enhancement techniques are applied, the process is expedited, and the freezing time is reduced to 175.09 seconds. To further analyze the solidification process, three time stages were selected to illustrate the movement of the solid front under different conditions of $Rd$ and $\gamma$. As time progresses, the solid front moves inward due to the dual cold regions. The movement of the solid front is faster as the value of radiation increases, indicating that radiation cooling enhances the freezing process. Additionally, the attendance of foam further hastens the movement of the solid front, demonstrating the effectiveness of combining these techniques to improve the solidification rate.

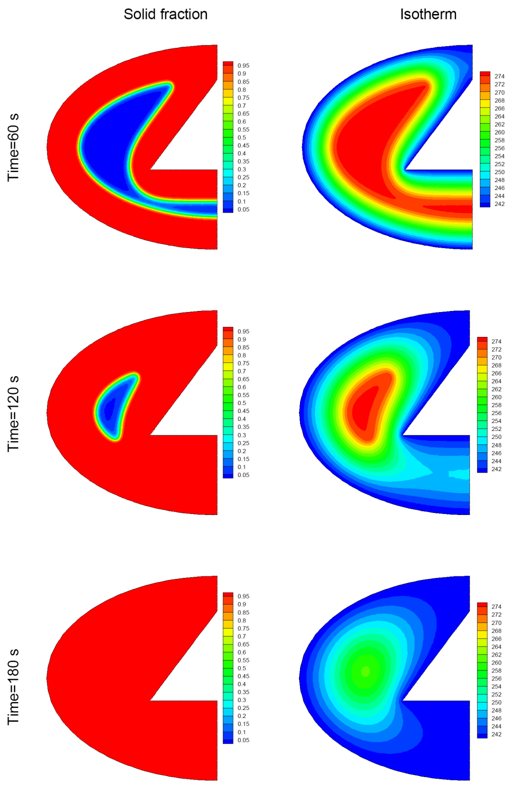


**Figure 6.** Changes in contours during freezing for the case of $\phi_{\mathrm{hnf}}$ = 0.02, $Rd$ = 1, $\gamma$ = 0.95

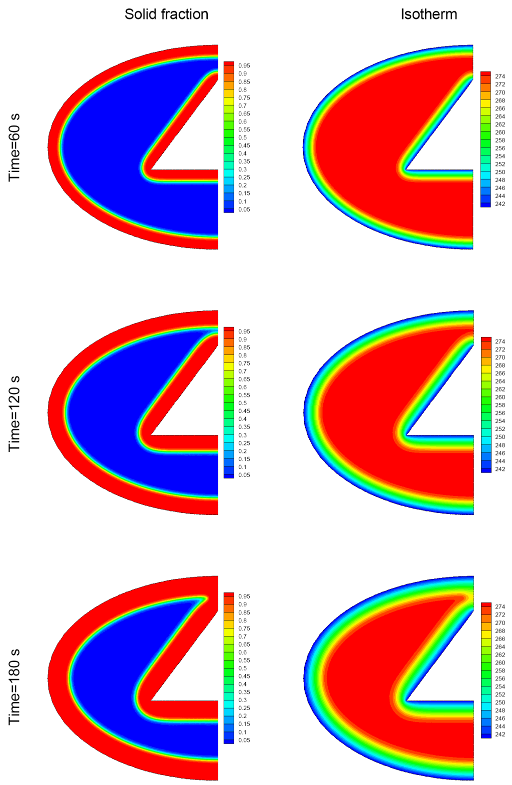


**Figure 7.** Changes in contours during freezing for the case of $\phi_{\mathrm{hnf}}$ = 0.02, $Rd$ = 0, $\gamma$ = 1

Note: $\phi_{\mathrm{hnf}}$ is the hybrid nanoparticle total volume fraction; $\gamma$ is the porosity of the porous metal foam; $Rd$ = the radiation parameter.

Figure 9, Figure 10, and Figure 11 present the evolution of three key factors over time. As time progresses, the domain gradually cools, and the decline in the liquid phase results in a corresponding decrease in the amount of energy present in the system. This decrease results directly from the liquid-to-solid phase transition of the PCM. As the solidification process continues, the solid fraction grows, and this trend is further accelerated by the presence of metal foam. The existence of foam not only speeds up the solidification process but also enhances the rate at which the solid fraction increases over time. The temperature reduction within the domain becomes more pronounced when porous foam and hybrid nanoparticles are used. The cooling effect is significantly enhanced, highlighting the importance of these techniques in controlling the thermal behavior. Besides, the introduction of thermal radiation further contributes to a decrease in the domain's temperature. Radiative cooling plays a key role in maintaining a colder environment, which accelerates the freezing process and promotes more efficient solidification.

Figure 12 illustrates the effect of the three enhancement techniques—porous foam, hybrid nanofluids, and radiative cooling—on the total time required for complete freezing. Among these techniques, the introduction of metal foam has the most noteworthy effect, reducing the solidification time by about 75.25%. This increase in the freezing speed is 2.19 times greater than the effect of radiation cooling and 14.5 times more than the impact of hybrid nanoparticles. Replacing water with hybrid nanofluids results in a smaller, but still noticeable, reduction in freezing time, approximately 5.19%. The incorporation of radiative cooling leads to a cooler domain, which helps expedite the freezing. The solidification time decreases by about 34.35% when radiative cooling is applied. Moreover, the impact of radiative cooling on the process is about 6.62 times more significant than the influence of hybrid nanoparticles, underlining the critical role of temperature control in speeding up solidification. When all three techniques are employed together, the freezing time decreases dramatically by 84.59%. This combined approach offers the most efficient solution for accelerating the solidification process, highlighting the potential of these integrated techniques to significantly enhance cold energy storage systems. The outputs determine that while each of the individual techniques offers improvements, the most substantial reduction in solidification time occurs when porous foam is utilized.

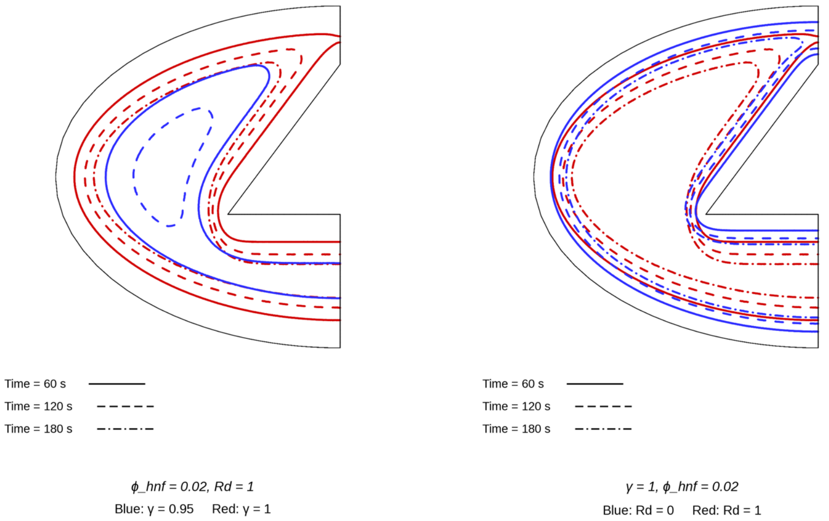


**Figure 8.** Comparison of ice-front evolution under different freezing conditions

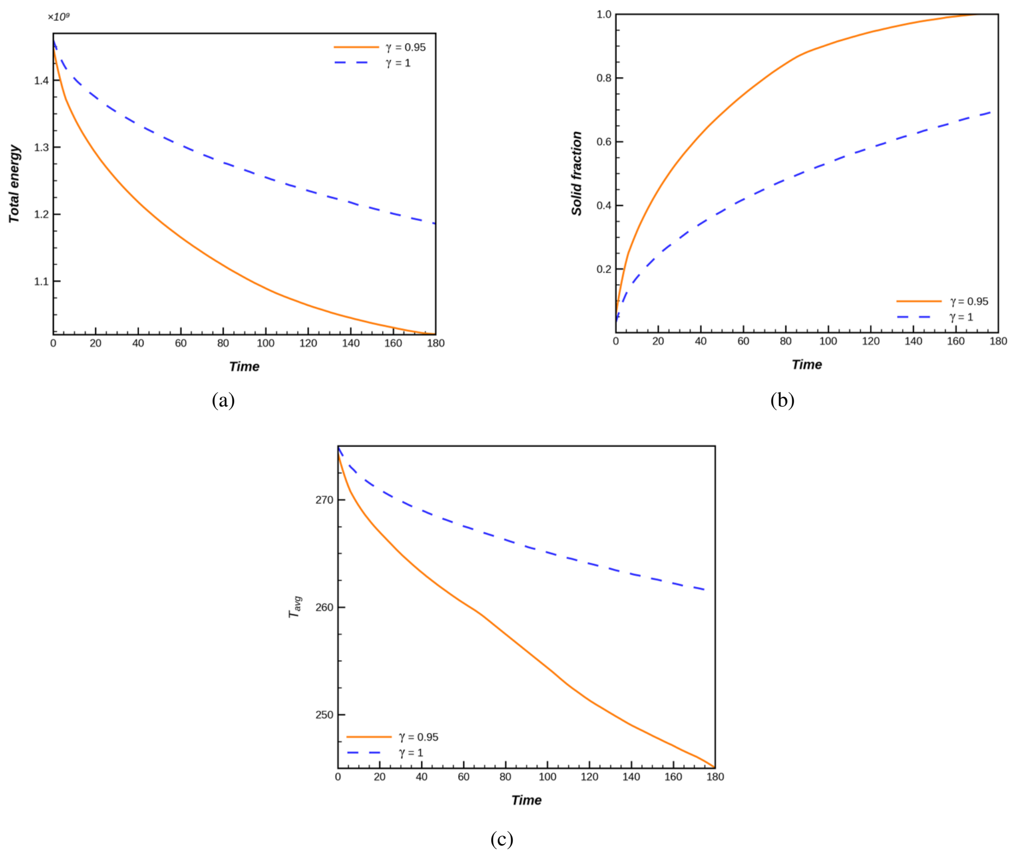


**Figure 9.** Temporal evolution of the monitored parameters for different values of the porosity of the porous metal foam $\gamma$

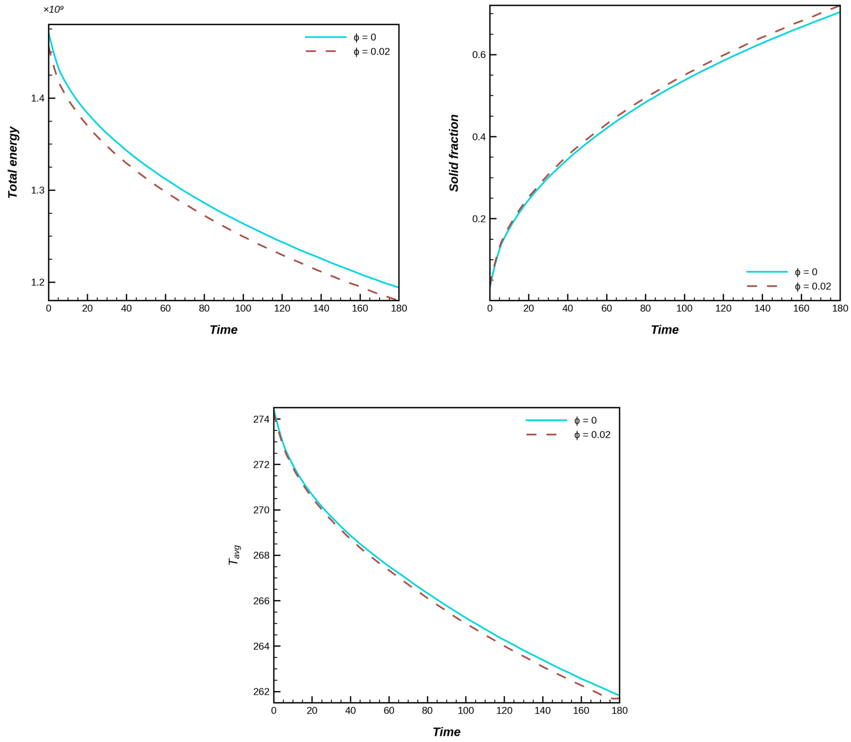


**Figure 10.** Temporal evolution of the monitored parameters for different values of the hybrid nanoparticle total volume fraction $\phi_{\mathrm{hnf}}$

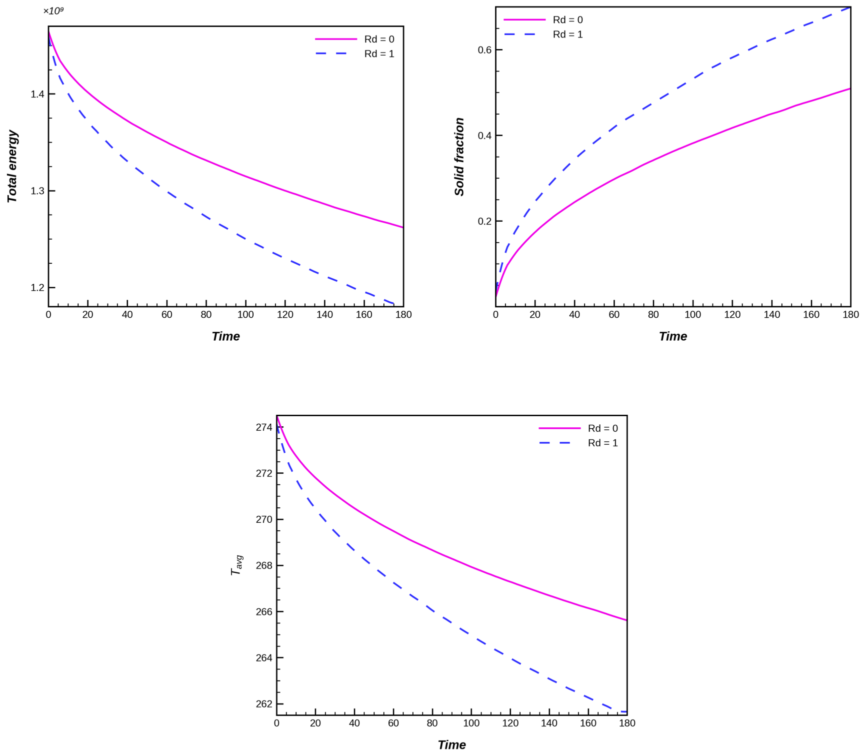


**Figure 11.** Temporal evolution of the monitored parameters for different values of the radiation parameter $Rd$

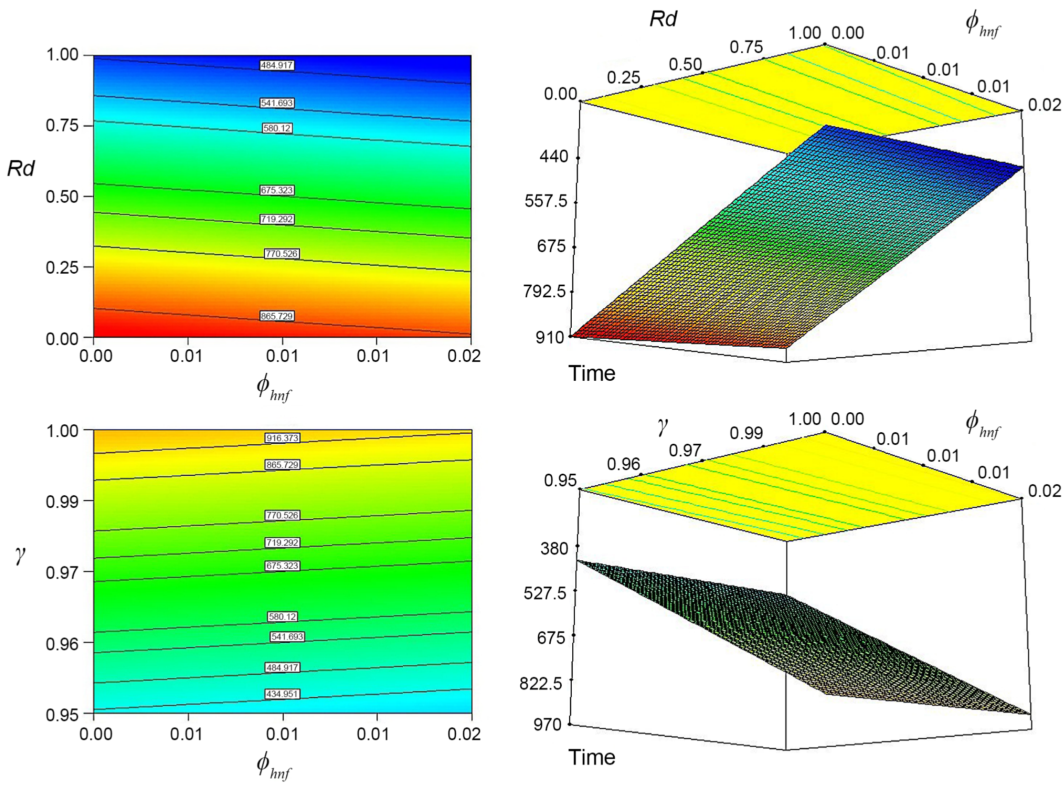


**Figure 12.** Total freezing time

Note: $\phi_{\mathrm{hnf}}$ is the hybrid nanoparticle total volume fraction; $\gamma$ is the porosity of the porous metal foam; $Rd$ = the radiation parameter.

## 4 Conclusion

In this study, a new container design featuring both triangular and elliptical cold surfaces was explored for cold energy storage applications. Instead of using traditional water, the container was filled with a hybrid nanomaterial, and porous foam was integrated to boost the freezing rate. The governing equations for this system were derived by assuming a homogeneous nanofluid mixture and neglecting the effect of gravity. The porosity of the foam was accounted for in determining the properties of the porous media. The model was simulated via the Galerkin method, known for its effectiveness in managing complex geometries. To evaluate the impact of the three key techniques—hybrid nanomaterials, porous foam, and the unique container shape—contour plots were generated to visualize the changes in temperature and solid fraction over time. A comparative analysis of different cases, focusing on the progression of the solid front, average scalar values, and freezing completion time, highlighted the significant improvements in freezing efficiency. The combination of metal foam and hybrid nanomaterials reduced the freezing time considerably, demonstrating the potential of this innovative approach for enhancing cold storage units.

This study advances current understanding by integrating multiple techniques in a novel way. As time progresses, the solid front moves inward due to the presence of two cold regions on both sides. This progression results in an increase in the solid fraction and a decrease in temperature. Notably, the areas near the elliptical and triangular surfaces solidify more rapidly than the regions farther away from these surfaces. When all three enhancement techniques—hybrid nanofluids, radiative cooling, and porous foam—are applied, the freezing time is meaningfully reduced by approximately 84.59%. Among these techniques, porous foam has the most substantial impact, leading to a decrement in freezing time by about 34.35%. The influence of radiative cooling on the process is found to be 6.62 times more significant than that of the nanoparticle concentration. The replacement of water with hybrid nanofluids leads to a 5.19% decrease in freezing time. The introduction of porous foam, however, has the most profound effect, reducing the required solidification time by a substantial 75.25%. This speed enhancement is 2.19 times and 14.5 times greater than the effects of radiative cooling and hybrid nanoparticles, respectively. The results indicate that porous foam plays a dominant role in improving the freezing speed, while the impact of hybrid nanoparticles is comparatively smaller. The phenomenon that becomes more pronounced when porous foam and hybrid nanoparticles are used in conjunction. As solidification progresses, the solid fraction increases, which is more prominent when porous foam is included in the system. Among all the scenarios tested, the longest solidification process occurs in the absence of radiation, taking approximately 1136.82 seconds. In contrast, when all three techniques are employed, the process is completed in just 175.09 seconds, demonstrating a significant improvement in freezing speed.

In conclusion, this study emphasizes the substantial benefits of integrating multiple techniques—porous foam, radiative cooling, and hybrid nanofluids—into cold energy storage systems. These methods not only accelerate the solidification process but also optimize the performance of PCMs in energy storage. The findings highlight the potential of these combined enhancements to achieve faster and more efficient thermal management solutions, which can lead to the advance of more effective and sustainable cold storage units.

**Data Availability**

The data used to support the research findings are available from the corresponding author upon request.

**Conflicts of Interest**

The author declares no conflicts of interest.

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